r/artificial

How much energy does AI use compared to humans?

Posted by u/NuseAI • • 0 points • 39 comments

- A recent paper challenges assumptions about the energy use of AI models, finding that AI systems emit significantly fewer carbon dioxide equivalents (CO_2e) compared to humans when producing text or images.

- The authors emphasize the importance of measuring carbon emissions from AI activities to inform sustainability policies.

- The ongoing debate among AI researchers highlights the challenges of accounting for the interactions between climate, society, and technology.

Source : https://venturebeat.com/ai/how-much-energy-does-ai-use-compared-to-humans-surprising-study-ignites-controversy/

Comments

u/[deleted] • 1 points • 2025-01-23 02:07 UTC

[removed]

u/soumyag101 • 1 points • 2024-03-04 11:02 UTC

"if Google replaced its current search algorithms with LLMs, SemiAnalysis estimates that each search request would cost up to 8.9 Wh of energy. Given Google's 9 billion search requests per day this would lead to an annual energy use of 29.2 terawatt-hours (TWh) on their servers, equivalent to the entire annual energy consumption of Ireland."

Save: [Exponential View](https://www.exponentialview.co/p/llm-search-and-energy-turing-transformation)

u/Extension-Copy-169 • 1 points • 2023-09-26 19:13 UTC

If we look at this from the background of
Energy landscape, it becomes more acute. Scaling renewable power has real limits.
Building enough wind, solar, and storage requires huge amounts of metals and minerals.
Starting new mines and reaching full production takes very, very long.
Relationships with countries controlling these resources stay uncertain.
Underinvesting in fossil fuels during the green shift caused supply-demand mismatches.

Fossil fuels remain critical in transition.
Nuclear still faces stigma.
Perfectly good nuclear plants get shut down early in Germany, even burning wood instead.
Same in California, without the wood burning.
Exceptions like Canada are rare.
The challenges to affordable, reliable, clean energy and quality water requirement for AI’s required scale are big

u/Enzor • 1 points • 2023-09-26 13:39 UTC

Anyone have a link to the arXiv.org study mentioned in the article? I've searched arXiv and see no such research study.

u/Holyragumuffin • 1 points • 2023-09-25 05:58 UTC

At present a helluva lot more. The brain uses as much power as a light bulb.

The energy-hungry nature of modern AI poses major bottlenecks to growing the models rapidly. At some point, humans will have to figure out low-power tricks, potentially inspired by biological neurons, to scale faster than Moore's law.

u/ButterscotchNo7634 • 2 points • 2023-09-25 05:28 UTC

That's a good question. We have to produce the electric energy for AI first and we should go backwards in the time. For example, Energy =100 W. How much energy do we have to use to produce 100 W. May-be 200 W. We do not use battery, Ai is taking energy from the line. Then we have to calculate CO2 we made for 200W. and so on. I would like to read a scientific study about it, but this article looks weak to me and is missing the data. Simply B.S.

u/water_bottle_goggles • 2 points • 2023-09-25 05:17 UTC

Way more